

1) $\rightarrow u_1 \leftarrow u_2$ $\cos \theta = 1$
 $\rightarrow v_1 \rightarrow v_2$ $\cos \theta = 1$

Impulse in direction of motion

$$v_2 - v_1 = e(u_1 + u_2) \quad (5)$$

$$I = \Delta mu$$

$$\overrightarrow{A} \quad -I = m(v_1 - u_1) \quad (5)$$

$$\overrightarrow{B} \quad I = m(v_2 + u_2) \quad (5)$$

$$8 \rightarrow \Delta \text{ kinetic energy} = \frac{1}{2} m(u_1^2 + u_2^2 - v_1^2 - v_2^2) \quad (5)$$

$$= \frac{1}{2} m [(u_1 + v_1)(u_1 + v_1) + (u_2 - v_2)(u_2 + v_2)]$$

$$= \frac{1}{2} I [u_1 + v_1 + u_2 - v_2]$$

$$= \frac{1}{2} I [(u_1 + u_2) + e(u_1 + u_2)] \quad (5)$$

$$= \frac{1}{2} I u (1 + e)$$

$$\text{Given } u = u_1 + u_2$$

[25]

2) $R = \frac{u^2}{g} \sin 2\alpha$

$H = \frac{u^2}{2g} \sin^2 \alpha$ } (5)

$$R = 2H$$

$$\frac{u^2}{g} \sin 2\alpha = \frac{u^2}{g} \sin^2 \alpha \quad (5)$$

$$2 \sin \alpha \cos \alpha = \sin^2 \alpha$$

$$2 = \tan \alpha$$

$$\alpha = \tan^{-1}(2) \quad (5)$$

$$R = \frac{u^2}{g} \sin 2\alpha$$

$$= \frac{u^2}{g} \frac{2 \tan \alpha}{1 + \tan^2 \alpha} \quad (5)$$

$$= \frac{u^2}{g} \frac{2 \times 2}{1 + 4}$$

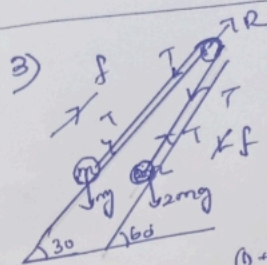
$$= \frac{4u^2}{5g} \quad (5)$$

[25]

$$l = a \sin \theta \times \cos \theta \times \frac{1}{\sin \theta \sqrt{4 + \tan^2 \theta}} = a \cos \theta \sqrt{4 + \tan^2 \theta}$$

(5)

$$= a \sqrt{1 + 3 \tan^2 \theta} \quad (5)$$



$$T - mg \sin 30 = m a \quad (5)$$

$$2mg \sin 60 - T = 2m a \quad (5)$$

$$(1) + (2) \quad \left(\frac{\sqrt{3}-1}{2} \right) mg = 2m a$$

$$a = \left(\frac{\sqrt{3}-1}{4} \right) g \quad (5)$$

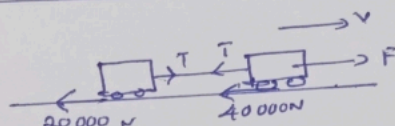
$$T = \frac{mg}{4} (\sqrt{3} + 1) \quad (5)$$

$$R = 2T \cos 15^\circ$$

$$= \frac{mg}{2} (\sqrt{3} + 1) \cos 15^\circ$$

[25] (5)

4)



$$F = ma$$

$$F - 60000 = m \cdot a \quad (5)$$

$$F = 60000 \quad (5)$$

$$P = Fv$$

$$900 \times 10^3 = 60 \times 10^3 v$$

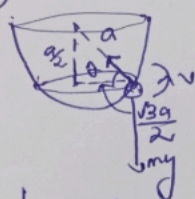
$$v = \frac{900}{60} = 15 \text{ m/s} \quad (5)$$

$$= 15 \text{ m/s} \quad (5)$$

$$= 54 \text{ km/h} \quad (5)$$

[25]

5)



$$R \sin \theta = mg \quad (5)$$

$$R \cos \theta = m \frac{v^2}{r} \quad (5)$$

$$\tan \theta = \frac{gr}{v^2}$$

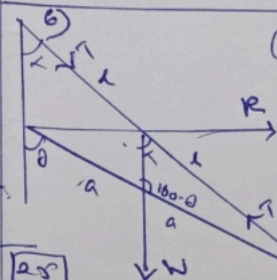
$$v^2 = \frac{gr}{\tan \theta} \quad (5)$$

$$= \frac{\sqrt{3}g \cdot \sqrt{3}g}{2}$$

$$v^2 = \frac{3g^2}{2}$$

$$v = \frac{1}{2} \sqrt{6ga} \quad (5)$$

[25]



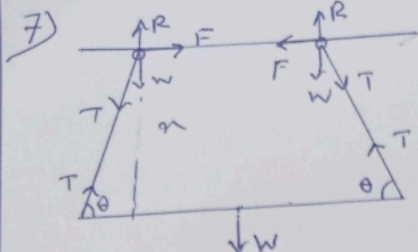
$$(1 + 1) \cos(180 - \theta) = 1 \cos 90 - 1 \cos \theta \quad (10)$$

$$\tan \alpha = \left(\frac{\tan \theta}{2} \right)$$

$$l \sin \alpha = a \sin \theta \quad (5)$$

$$l = \frac{a \sin \theta}{\sin \alpha}$$

$$= \frac{a \sin \theta}{\tan \alpha \sqrt{4 + \tan^2 \alpha}}$$



$$\begin{aligned} \uparrow 2T \sin \theta &= W \\ 2R &= 3W \\ R &= \frac{3W}{2} \end{aligned} \quad (5)$$

$$\rightarrow T \cos \theta = F$$

$$\frac{2T \sin \theta}{T \cos \theta} = \frac{W}{F}$$

$$2 \tan \theta = \frac{W}{F} \quad (5)$$

$$\frac{F}{R} \leq M$$

$$\frac{\frac{W}{2 \tan \theta}}{\frac{3W}{2}} \leq M$$

$$\cot \theta \leq 3M \quad (5)$$

$$2.50 \sin \theta \leq 1.59 \text{ radians} \quad (5)$$

$$\sin \theta \leq \frac{3}{5}$$



$$\cot \theta \geq \frac{4}{3}$$

$$\therefore \frac{4}{3} \leq 3M$$

$$M \geq \frac{4}{9} \quad (5)$$

25

$$08) \quad a + 3b \leq 7a - 5b \quad (5)$$

$$\therefore (a + 3b) - (7a - 5b) \geq 0 \quad (5) \quad (1)$$

$$7a^2 - 15ab + 16a \cdot b \geq 0$$

$$(a - 5b) \cdot (7a + 3b) = 0 \quad (5)$$

$$7a^2 - 15ab + 16a \cdot b = 0 \quad (5) \quad (2)$$

$$\begin{aligned} 1) - 2) \quad 48ab &= 0 \quad (5) \\ ab &= 0 \end{aligned}$$

a or b is zero

25

9) RAUBUL

$$= P_A + P_B + P_C - P_{AB} - P_{BC} - P_{AC} + P_{ABC} \quad (5)$$

$$= 0.3 + 0.4 + 0.8 - 0.08 - 0.28 - 0.09 + 0.09 \quad (5)$$

$$1.23 - n \geq 0.75 \quad (5)$$

$$0.23 \leq n \leq 0.48 \quad (5)$$

25

10) $a + a + d, \dots, a + 2nd$ terms

$$\bar{n} = \frac{(2n+1)[a + a + 2nd]}{2} \quad (5)$$

$$= \frac{(2n+1)[a + nd]}{(2n+1)}$$

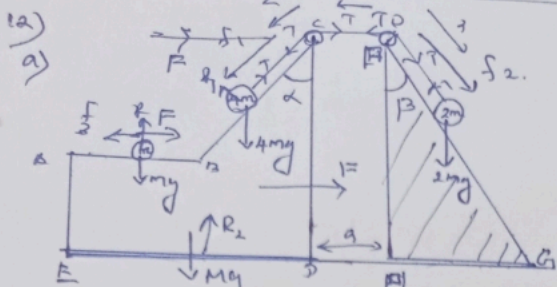
$$= a + nd \quad (5)$$

$$\text{Sum} = \frac{nd + (n-1)d + (n-2)d + \dots + 0 + d + nd}{2n+1} \quad (10)$$

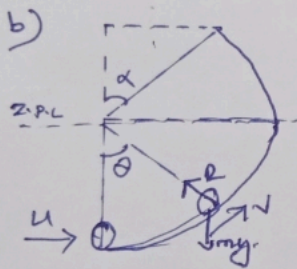
$$= \frac{n[nd + d] \times 2}{(2n+1)} \quad (5)$$

$$= \frac{n(n+1)d}{(2n+1)}$$

25

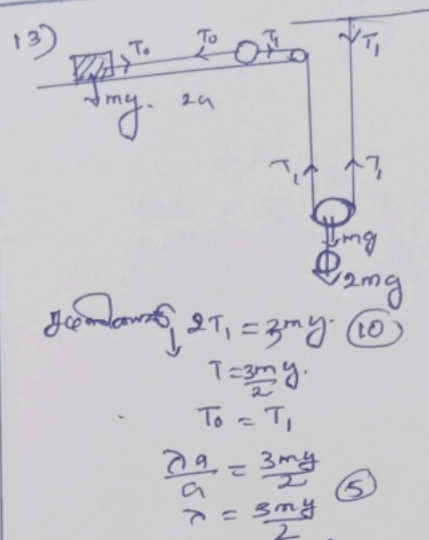


$x + y + z = \text{constant}$
 $\ddot{x} + \ddot{y} + \ddot{z} = 0$
 $f_2 - F + f_1 = 0$ (10)
 $\odot \rightarrow 0 = m(\ddot{y} - F)$ (2)
 $\odot \rightarrow K \quad 4my \cos \alpha - T = 4m(F - F \sin \alpha)$ (3)
 $\odot \rightarrow T = MF + m(F - f_2) + 4m(F - f_1 \sin \alpha)$ (15)
 $\odot \rightarrow 2mg \cos \beta - T = 2m \ddot{z}$ (20)
 $\odot \rightarrow 2mg \cos \beta - T = 2m \ddot{z}$ (5)
85



$\frac{1}{2} m u^2 - m g a = \frac{1}{2} m v^2 - m g a \cos \theta$ (20)
 $u^2 - 2ga = v^2 - 2ga \cos \theta$
 $v^2 = u^2 - 2ga(1 - \cos \theta)$ (5) (1) 28
 $T = F = ma$
 $R - mg \cos \theta = m \frac{v^2}{a}$ (10)
 $R = mg \cos \theta + \frac{m}{a} (u^2 - 2ga(1 - \cos \theta))$ (15)
 $\Rightarrow \ddot{z} + \frac{dv}{dt} = 0 - 2ga(1 + \sin \theta) \frac{d\theta}{dt}$
 $v \frac{dv}{dt} = -2ga \sin \theta \frac{d\theta}{dt}$ (10)
 $v = a \dot{\theta} \quad \frac{dv}{dt} = a \ddot{\theta}$
 $a \dot{\theta} \cdot a \ddot{\theta} = -ga \sin \theta a \dot{\theta}$ (15)
 $a \ddot{\theta} = -ga \sin \theta$ (5)
 $\therefore \ddot{\theta} = -ga \sin \theta$

$\theta = \pi - \alpha$
 $v^2 = u^2 - 2ga(1 - \cos(\pi - \alpha))$ (5)
 $= u^2 - 2ga(1 + \cos \alpha)$
 $= u^2 - \frac{14ga}{4}$ (5) 10
 $= u^2 - \frac{7ga}{4}$ 150



$2y + z = \text{constant}$
 $x + z = \text{constant}$
 $\therefore 2\ddot{y} + \ddot{z} = 0$ & $\ddot{x} + \ddot{z} = 0$
 $2\ddot{y} = -\ddot{z}$ & $\ddot{x} - 2\ddot{y} = 0$ (10)
 $\ddot{x} = 2\ddot{y}$ (10)
 $\odot \rightarrow -T + T_2 = m \ddot{x}$ (10)
 $\odot \downarrow \quad my - 2T_2 = m \ddot{y}$ (10)
 $my - 2T = \frac{5m \ddot{y}}{2}$
 $my - 2 \cdot \frac{3mg}{2} (1 - a) = \frac{5m \ddot{y}}{2}$ (10)
 $my \left[1 - \frac{3(1-a)}{a} \right] = \frac{5m \ddot{y}}{2}$
 $\ddot{y} = \frac{2g}{5a} [4a - 3n]$
 $\ddot{n} = -\frac{6g}{5a} (n - \frac{4a}{3})$ (10) 60
 $\ddot{n} + \frac{6g}{5a} (n - \frac{4a}{3}) = 0$

13) ஒரு வட்டம்: - - - - -

$$P = \pi - \frac{4g}{3} \text{ மீட்டர்கள்}$$

$$\dot{P} = \dot{\pi}$$

$$\ddot{P} = \ddot{\pi}$$

$$\ddot{P} = -\frac{6g}{5a} (P) \text{ மீட்டர்கள்/செக்யூரில்}$$

∴ இக்கோணம் இக்கோணம் S.H.M ஆகும். (10)

சமன்பாடு $\ddot{\pi} = C$

$$\pi - \frac{4g}{3} = C$$

$$\pi = \frac{4g}{3} \quad (10)$$

ஒவ்வொரு $\frac{4g}{3}$ மீட்டர்கள் சமன்பாடு

$$2\pi = 2 \times \frac{4g}{3} = \frac{2g}{3} \quad (10)$$

சமன்பாடு $\ddot{\pi} = C$ மீட்டர்கள்

2 மீட்டர்கள் $\frac{2g}{3}$ மீட்டர்கள்

இக்கோணம் இக்கோணம் $\pi = \frac{4g}{3}$ மீட்டர்கள்

இக்கோணம் இக்கோணம்

இக்கோணம் இக்கோணம்

இக்கோணம் இக்கோணம்

$\pi = \frac{4g}{3}$ மீட்டர்கள் $\pi = \frac{4g}{3}$ மீட்டர்கள்

$$\sqrt{\frac{2}{3}} \omega (\frac{4g}{3} - \frac{4g}{3})$$

$$\sqrt{\frac{2}{3}} \omega (\frac{4g}{3} - \frac{4g}{3})$$

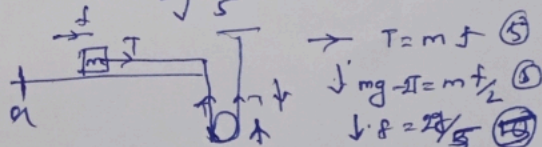
$$= \frac{3g}{4} \omega^2 \quad 5$$

$$\omega = \frac{\sqrt{3} g \omega}{3}$$

$$= \frac{\sqrt{3} g \omega}{3}$$

$$= \sqrt{\frac{18g}{5}}$$

$$= \sqrt{\frac{2g}{5}}$$

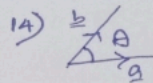


$$\textcircled{A} \leftarrow \sqrt{v^2 + \omega^2 a^2}$$

$$0 = \frac{2g}{5} - \frac{2g}{5} \quad (5)$$

$$S = a \quad (5)$$

∴ இக்கோணம் இக்கோணம் $\pi = \frac{4g}{3}$ மீட்டர்கள்



$$a \cdot b = |a| |b| \cos \theta$$

இக்கோணம் $0 \leq \theta \leq \pi$, θ இக்கோணம் இக்கோணம்

$$OA \geq OB \geq OC \Rightarrow |a| \geq |b| \geq |c|$$

$$a + b + c = 0 \quad 5$$

$$a \cdot a + a \cdot b + a \cdot c = 0 \quad (5)$$

$$|a|^2 = a \cdot b + a \cdot c \quad (1) \quad (5)$$

$$|b|^2 = a \cdot b + c \cdot b \quad (2) \quad (5)$$

$$|c|^2 = a \cdot c + b \cdot c \quad (3) \quad (5)$$

$$|a|^2 = |b|^2 = |c|^2$$

$$\therefore a \cdot b = a \cdot c = b \cdot c \quad (5)$$

$$\vec{AB} = b - a \quad (5)$$

$$AB^2 = |\vec{AB}|^2 \quad (5)$$

$$= \vec{AB} \cdot \vec{AB}$$

$$= (b - a) \cdot (b - a)$$

$$AB^2 = |b|^2 + |a|^2 - 2a \cdot b$$

$$AB^2 = |a|^2 + |b|^2 - 2a \cdot b \quad (5)$$

$$2|a|^2 = AB^2 + 2a \cdot b \quad (5)$$

$$2|b|^2 = BC^2 + 2b \cdot c \quad (5)$$

$$2|c|^2 = AC^2 + 2c \cdot a \quad (5)$$

$$|a| = |b| = |c| \quad \& \quad a \cdot b = b \cdot c = a \cdot c$$

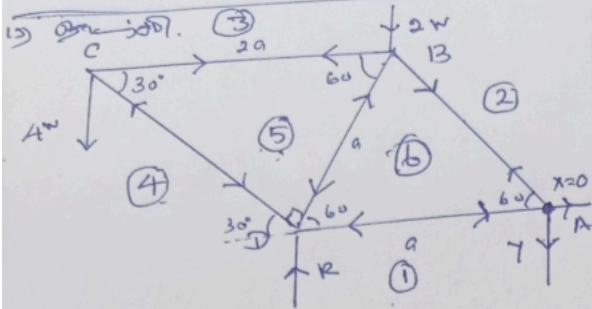
$$\therefore AB^2 = BC^2 = AC^2$$

$$AB = BC = AC$$

ABC மீட்டர்கள் $\pi = \frac{4g}{3}$ மீட்டர்கள்

மீட்டர்கள்





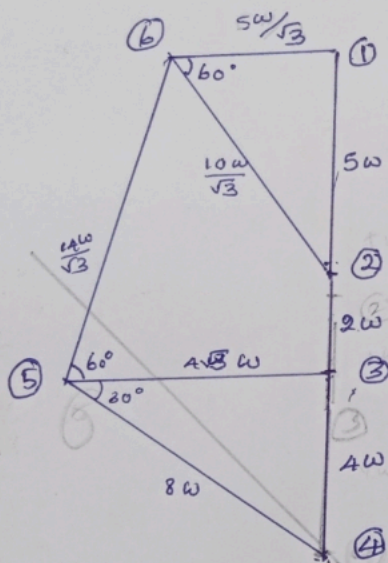
$$R = Y + 6W \quad (5)$$

$$D) \quad 4W \frac{3a}{2} - 2W \times \frac{a}{2} \Rightarrow Y a = 0 \quad (5)$$

$$6W - W = Y$$

$$Y = 5W$$

$$R = 11W$$



(30)

Members 2) Load 3) Reactions

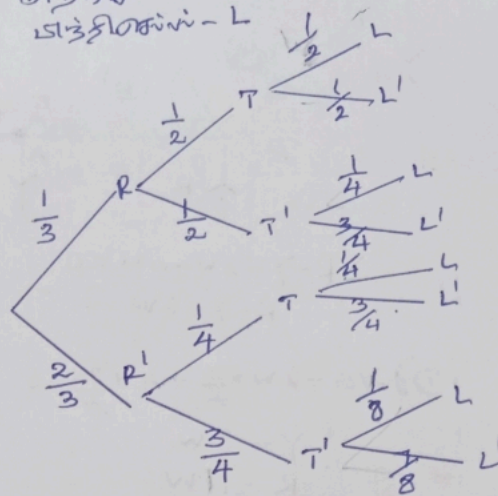
AB	$10W/\sqrt{3}$	-
BC	$4\sqrt{3}W$	-
CD	-	$8W$
DA	-	$5W/\sqrt{3}$
DB	-	$14W/\sqrt{3}$

(50)

90

150

17) Long span - R
Short span - T
Stiffness - L



$$1) \quad \frac{2}{3} \times \frac{1}{4} \times \frac{1}{4} \quad (10)$$

$$= \frac{2}{48}$$

$$= \frac{1}{24} \quad (5)$$

$$1) \quad \frac{1}{3} \times \frac{1}{2} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2} \times \frac{1}{4} + \frac{2}{3} \times \frac{1}{4} \times \frac{1}{4} + \frac{2}{3} \times \frac{3}{4} \times \frac{1}{8} \quad (5)$$

$$= \frac{1}{12} + \frac{1}{24} + \frac{2}{48} + \frac{6}{96}$$

$$= \frac{8 + 4 + 4 + 6}{96} \quad (8)$$

$$= \frac{22}{96} \quad (5)$$

$$= \frac{11}{48}$$

$$11) \quad \frac{\frac{1}{12} + \frac{1}{24}}{\frac{11}{48}} = \frac{6}{11} \quad (25)$$

17 Q. 5.57.

2 wgs	Len. of	m_i	$f_i m_i$	$f_i m_i^2$
120-130	2	125	250	31250
130-140	5	135	675	91125
140-150	25	145	3625	525625
150-160	10	155	1550	240250
160-170	8	165	1320	217800
	50	(5)	7420	1,106,050
			(10)	(10)

$$\text{Ans } \mu = \frac{7420}{50} = 148.4 \quad (5)$$

$$\text{Ans } \mu = L + \left(\frac{\frac{N}{2} - C.F.}{f} \right) C \quad (5)$$

$$= 140 + \left(\frac{25 - 7}{25} \right) 10 \quad (5)$$

$$= 147.2 \quad (5)$$

$$\text{Ans } \sigma = \sqrt{\frac{\sum f_i m_i^2}{\sum f_i} - \mu^2} \quad (5)$$

$$= \sqrt{\frac{1,106,050}{50} - 148.4^2} \quad (5)$$

$$= \sqrt{22121 - 22022.56} \quad (5)$$

$$= \sqrt{98.44} \quad (5)$$

$$= 9.97 \quad (5)$$

$$K = 3 \left(\frac{\mu - M}{\sigma} \right) = 3 \left(\frac{148.4 - 147.2}{9.97} \right) \quad (5)$$

$$= 0.36 \quad (5)$$

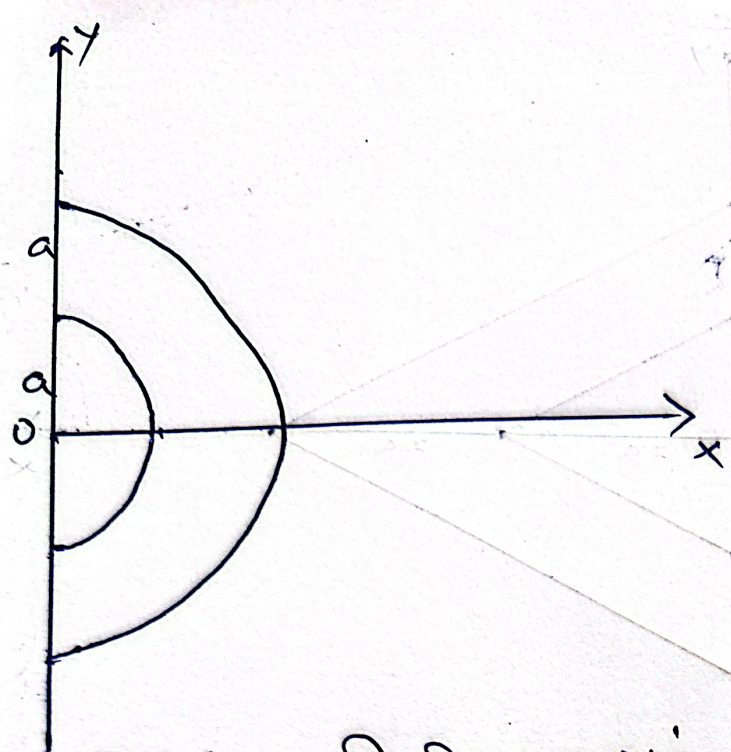
$$M_0 = \frac{7420 + 290}{52} = 148.2 \quad (10)$$

$$\sigma_0 = \sqrt{\frac{1,106,050 + 125^2 + 165^2}{52} - 148.2^2} \quad (10) \quad (5)$$

$$= \sqrt{130.9907} \quad (5)$$

$$= 11.45 \quad (5)$$

150



இந்த வளைந்த பகுதி திணிவு அடர்த்தி ρ உள்ளது. Ox அச்சின் கருத்து. (5)
 $y_2 = 0$

வளைவு	திணிவு	Oy அ.கி. திணிவு
	$\frac{2}{3} \pi a^3 \rho_2$ $= \frac{16}{3} \pi a^3 \rho_2$ (5)	$\frac{3 \cdot (2a)}{8}$ $= \frac{3a}{4}$ (5)
	$\frac{2}{3} \pi a^3 \rho_2$ (5)	$\frac{3a}{8}$ (5)
	$\frac{14}{3} \pi a^3 \rho_2$ (5)	$\frac{3a}{8}$

$$\frac{14}{3} \pi a^3 \rho_2 \cdot \bar{x}_2$$

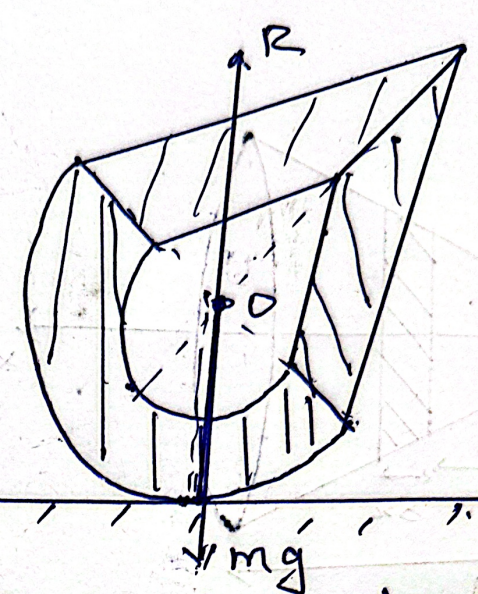
$$= \frac{16}{3} \pi a^3 \rho_2 \cdot \frac{3a}{4} - \frac{2}{3} \pi a^3 \rho_2 \cdot \frac{3a}{8}$$

(15)

$$\frac{14}{3} \bar{x}_2 = \frac{12}{3} a - \frac{1}{3} \cdot \frac{3a}{4}$$

$$\bar{x}_2 = \frac{45}{56} a$$

(5)



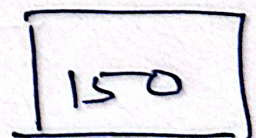
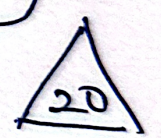
இந்த வளைந்த பகுதி திணிவு அடர்த்தி ρ உள்ளது. Ox அச்சின் கருத்து. (5)
 $y_2 = 0$

$$\frac{14 \pi a^3 \rho_1}{3} \cdot \frac{15a}{14} = \frac{14 \pi a^3 \rho_2}{3} \cdot \frac{45a}{56}$$

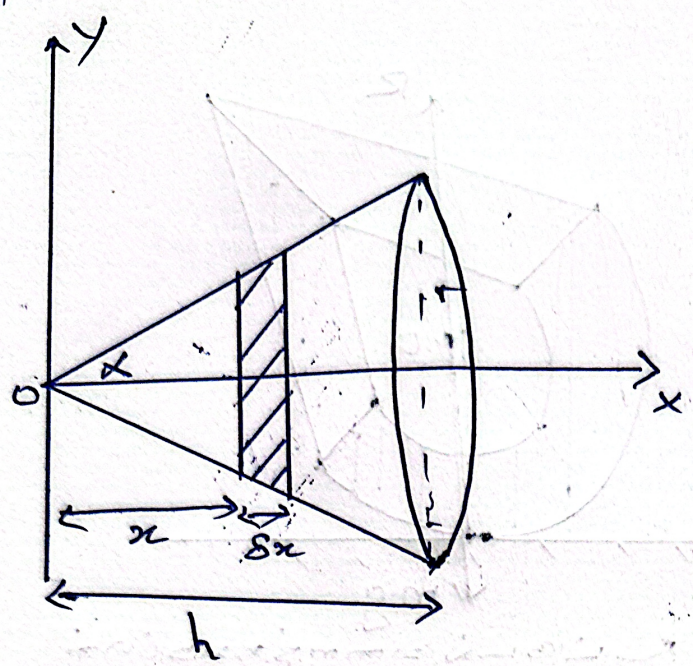
$$\rho_1 \cdot a = \rho_2 \cdot \frac{3a}{4}$$

$$\frac{\rho_1}{\rho_2} = \frac{3}{4}$$

(5)



16



$$\tan \alpha = \frac{r}{h}$$

$$r = h \tan \alpha$$

தடத்திவிட்டது திணிவு Ox அச்சின் கருக்கம், (5)

$$\bar{y} = 0$$

$$\bar{x} = \frac{\int_0^h \pi x^2 \tan^2 \alpha \cdot \rho \cdot x dx}{\int_0^h \pi x^2 \tan^2 \alpha \cdot \rho \cdot dx} \quad (5)$$

$$\int_0^h \pi x^2 \tan^2 \alpha \cdot \rho \cdot dx \quad (5)$$

$$= \frac{\frac{x^4}{4} \Big|_0^h}{\frac{x^3}{3} \Big|_0^h} \quad (5)$$

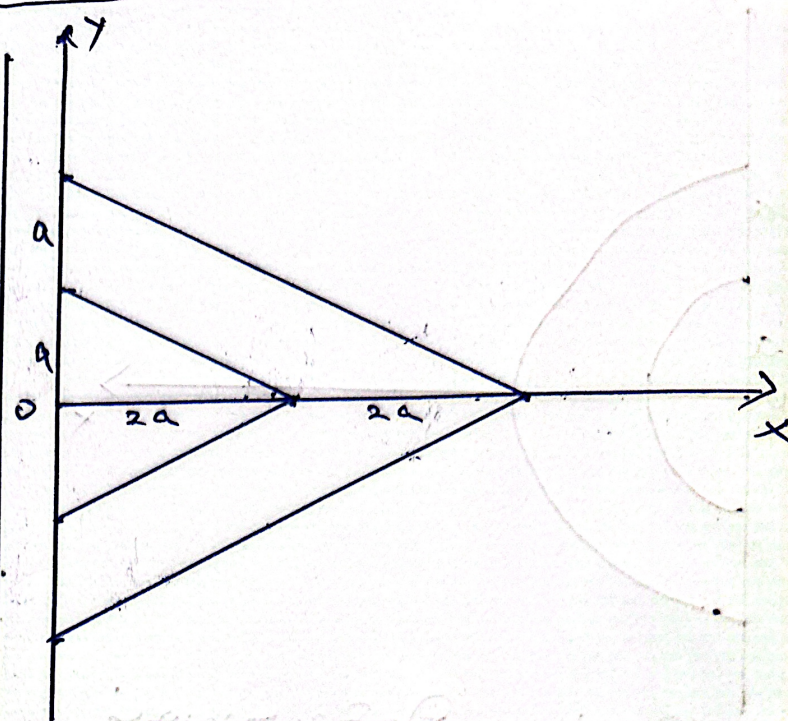
$$\frac{\frac{h^4}{4}}{\frac{h^3}{3}} \quad (5)$$

$$= \frac{\frac{h^4}{4}}{\frac{h^3}{3}}$$

$$= \frac{3h}{4}$$

30

$\therefore$ தடத்திவிட்ட திணிவுத் திணிவுத் தி. நம. தி. $= h - \frac{3h}{4} = \frac{h}{4}$ (5)



தடத்திவிட்டது திணிவு Ox அச்சின் கருக்கம், (5) $\bar{y}_1 = 0$

உரு	திணிவு	தடத்திவிட்ட திணிவுத் தி. நம. தி.
	$\frac{1}{3} \cdot \pi \cdot (2a)^2 \cdot 4\rho$ $= \frac{16}{3} \pi a^3 \rho$ (5)	a (5)
	$\frac{1}{3} \cdot \pi \cdot a^2 \cdot 2\rho$ $= \frac{2}{3} \pi a^3 \rho$ (5)	$\frac{a}{2}$ (5)
	$\frac{14}{3} \pi a^3 \rho$ (5)	$\bar{x}_1$

$$\frac{14}{3} \pi a^3 \rho \cdot \bar{x}_1 = \frac{16}{3} \pi a^3 \rho \cdot a - \frac{2}{3} \pi a^3 \rho \cdot \frac{a}{2} \quad (15)$$

$$\frac{14}{3} \cdot \bar{x}_1 = \frac{16}{3} \cdot a - \frac{1}{3} a$$

$$\bar{x}_1 = \frac{15}{14} a \quad (5)$$

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